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Unit 4 : Session 7 : SLO 2 | Odd Man Out : Generation of Control Signals

I. Odd Man Out

1. Group Opcode, Operand, Clock, Condition Code → Clock → a timing reference; the others are instruction/state information that feed control-signal generation.
2. Group Control Memory, ALU, Decoder, Flip-flops → ALU → a data-processing unit; the others are control-path elements.
3. Group MEMREAD, MEMWRITE, ALUOP, DATAPATH → DATAPATH → a hardware structure; the others are control signals.
4. Group Hardwired, Microprogrammed, Combinational, RISC → RISC → an ISA design style; the others are control-unit implementation approaches.
5. Group Sequence Counter, Program Counter, Instruction Register, Control Store → Control Store → exists only in microprogrammed units; the others are general processor registers/counters.
6. Group Horizontal Microcode, Vertical Microcode, Nanocode, Opcode → Opcode → an instruction field; the others are microprogram encoding formats.
7. Group AND gate, OR gate, Decoder, Cache Memory → Cache Memory → a storage structure; the others are logic elements.
8. Group Fetch, Decode, Execute, Compile → Compile → a software translation step; the others are hardware instruction-cycle stages.
9. Group Control Signal, Data Signal, Address Signal, Interrupt Signal → Interrupt Signal → an asynchronous event line; the others are the three standard bus signal groups.
10. Group Finite State Machine, Control Matrix, Pipeline Registers, Branch Predictor → Branch Predictor → a performance feature; the others are structures used to implement/generate control sequencing.